

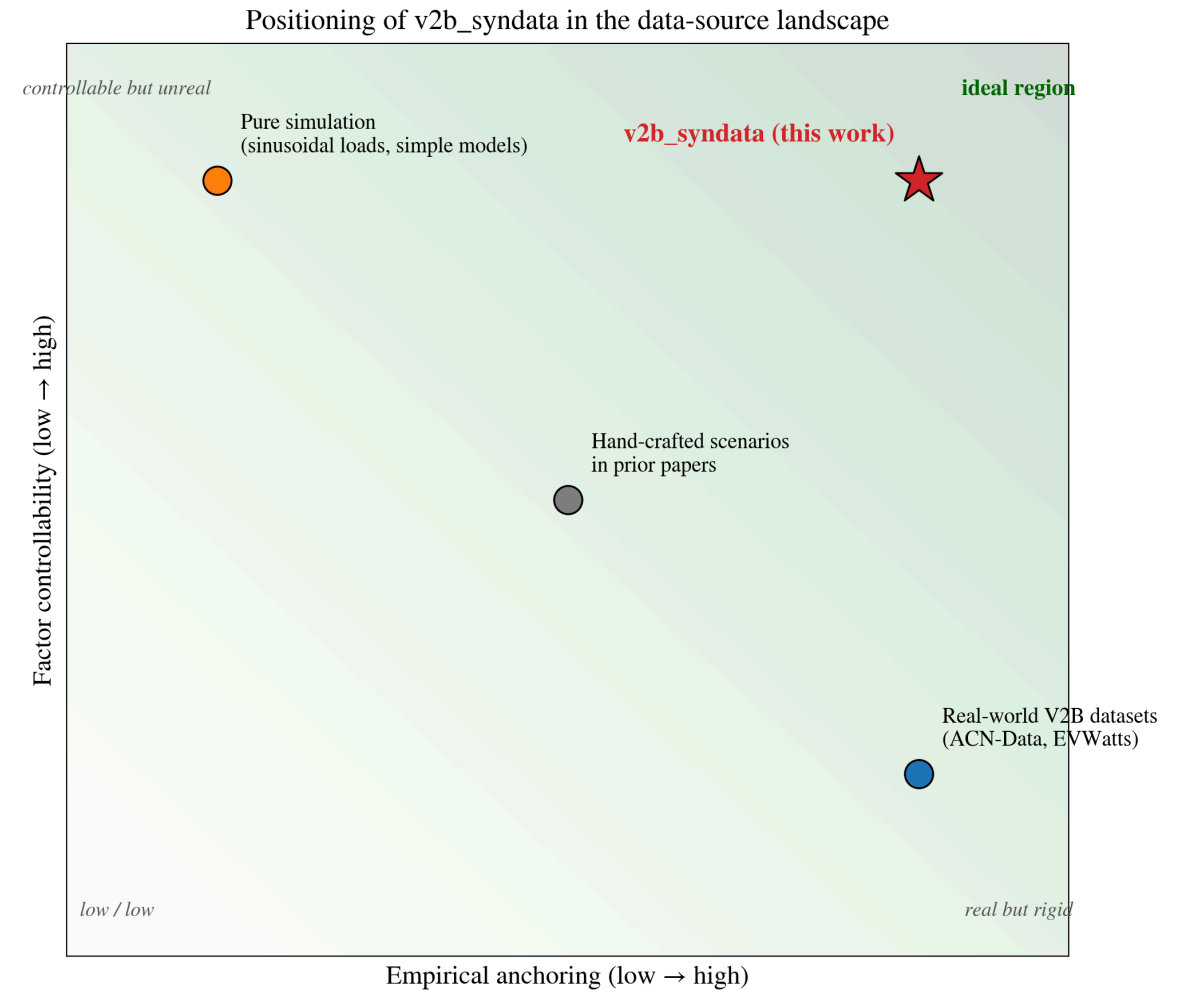
v2b_syndata

Synthetic data for Vehicle-to-Building research

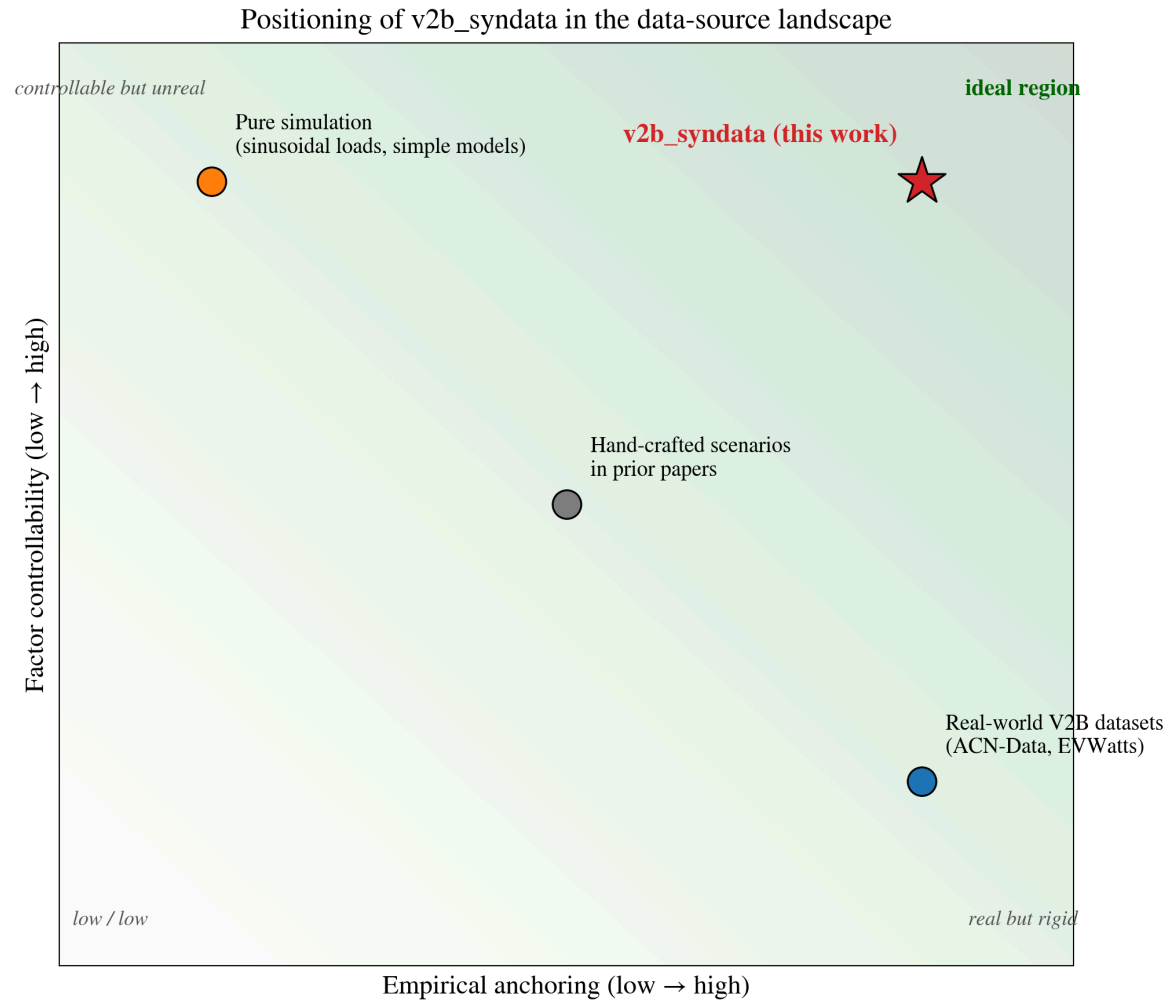
Parameter-driven, seed-reproducible, audited.

1. The problem

- V2B sits at the intersection of buildings, vehicles, users, equipment, tariffs, grid events.
- Empirical traces that span all six are rare or partial.
- Without a shared substrate, cross-study comparison is apples to oranges.



Position diagram



2. What it produces

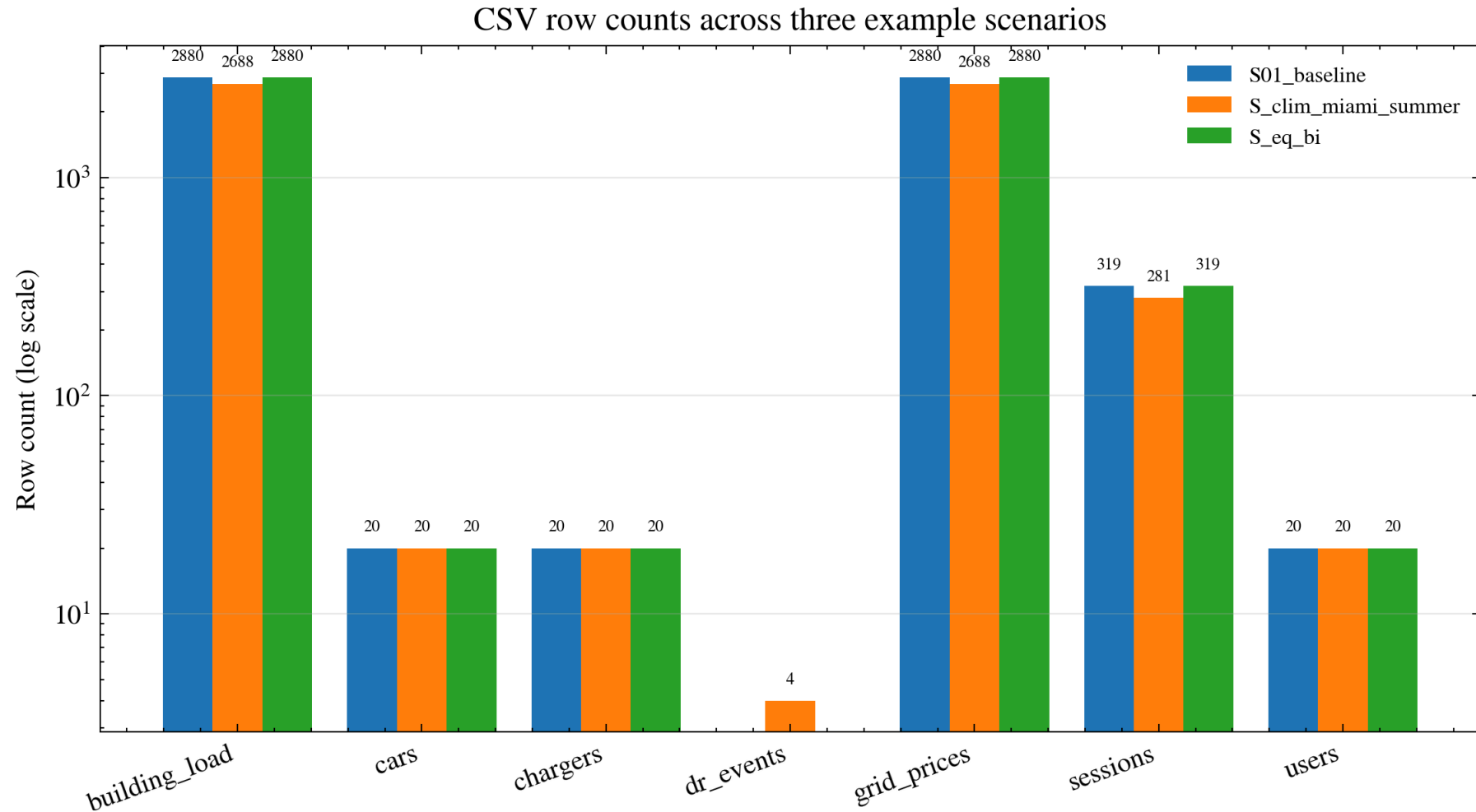
Seven CSVs + a `manifest.json` per scenario.

- `building_load.csv` , `cars.csv` , `users.csv`
- `chargers.csv` , `sessions.csv`
- `grid_prices.csv` , `dr_events.csv`
- Manifest pins all resolved knobs, seed lineage, per-CSV SHA-256.

CSV schema

CSV	Rows	Key columns
building_load.csv	~2700/mo	datetime, power_flex_kw, power_inflex_kw
cars.csv	20–50	car_id, capacity_kwh, battery_class
users.csv	20–50	car_id, region, phi, kappa, delta_km
chargers.csv	1–100	charger_id, directionality, max_rate_kw
sessions.csv	~300–800	car_id, arrival, departure, arrival_soc
grid_prices.csv	~2700/mo	datetime, price_per_kwh, type
dr_events.csv	0–N	event_id, start, end, magnitude_kw

Schemas at a glance



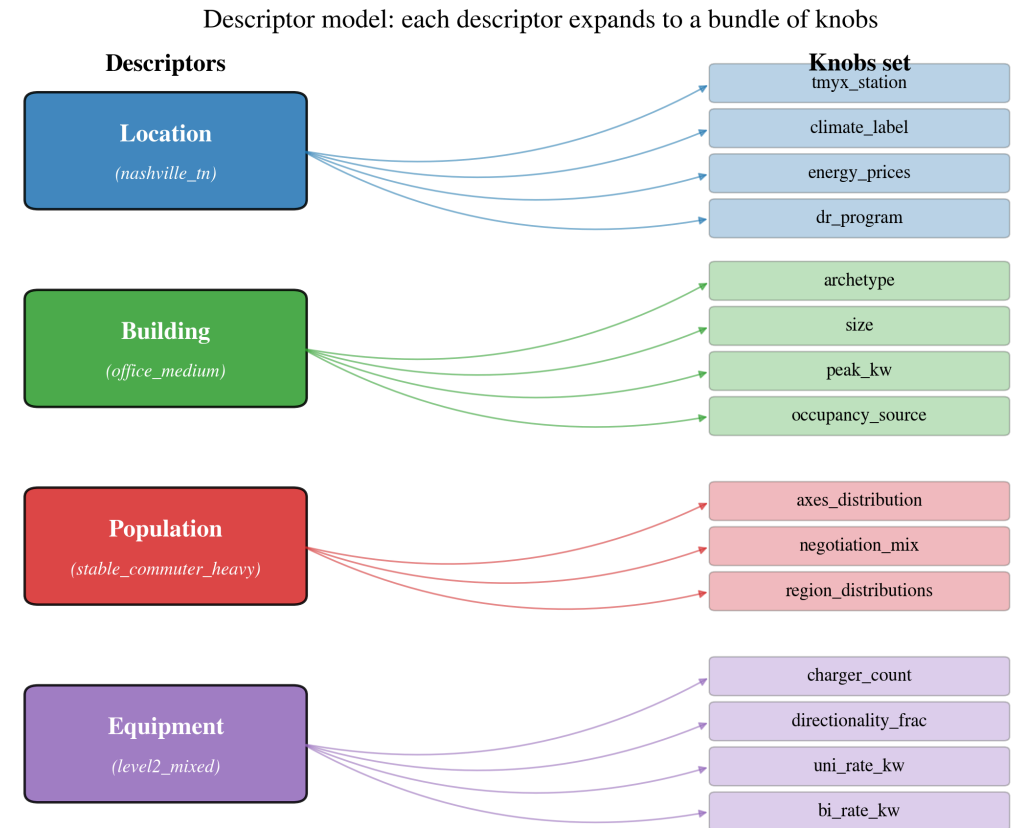
Manifest excerpt

manifest.json (excerpt) — S01_baseline

```
{
  "scenario_id": "S01",
  "seed": 42,
  "knob_resolution": {
    "building_load.archetype": {
      "source": "descriptor:medium_office_v1",
      "value": "office"
    },
    "sim_window.custom_end": {
      "source": "default",
      "value": null
    },
    "user_behavior.region_distributions.erratic.arrival.mu": {
      "source": "hand_specified:consent_default",
      "value": 11.0
    },
    "building_load.climate": {
      "source": "descriptor:nashville_tn",
      "value": "subtropical"
    }
  },
  "generator_git_sha": "3dd97fc94b5c5f36ff0dc1da8f7167247f4cb03b",
  "generated_at": "2026-05-19T22:55:43.098719Z",
  "csv_sha256": {
    "building_load": "718d65670d1704bb...c4618cef",
    "cars": "7a8c3e2ea392d455...414efa41",
    "chargers": "358213e771254b62...b6224735"
  }
}
```

3. The factor space

- Population: ψ (willingness), region mix, dispersion.
- Building, equipment, tariff, DR, climate.
- 98 knobs total, all in `configs/knobs.yaml`.



Knob buckets

98 tunable knobs grouped by bucket

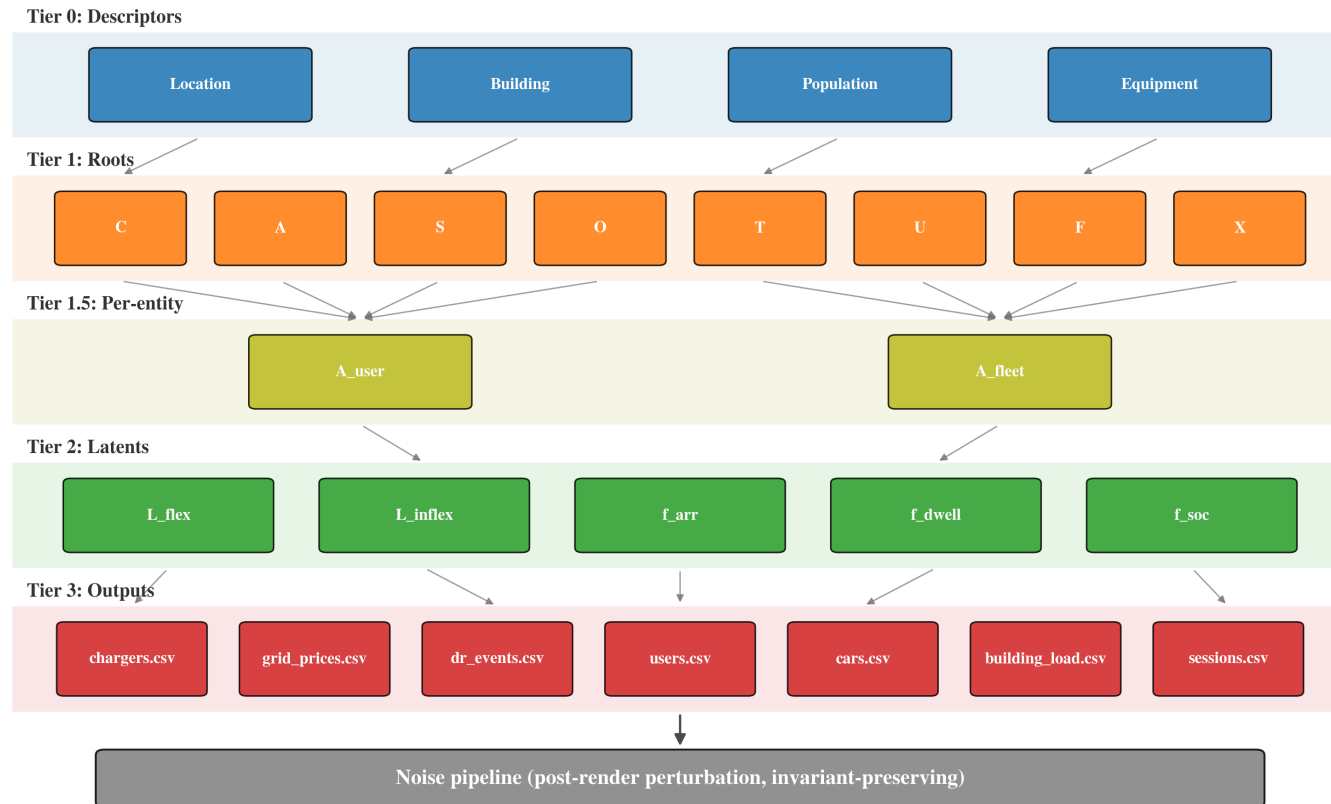


Experiment matrix

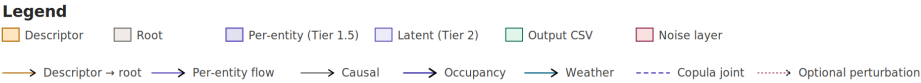
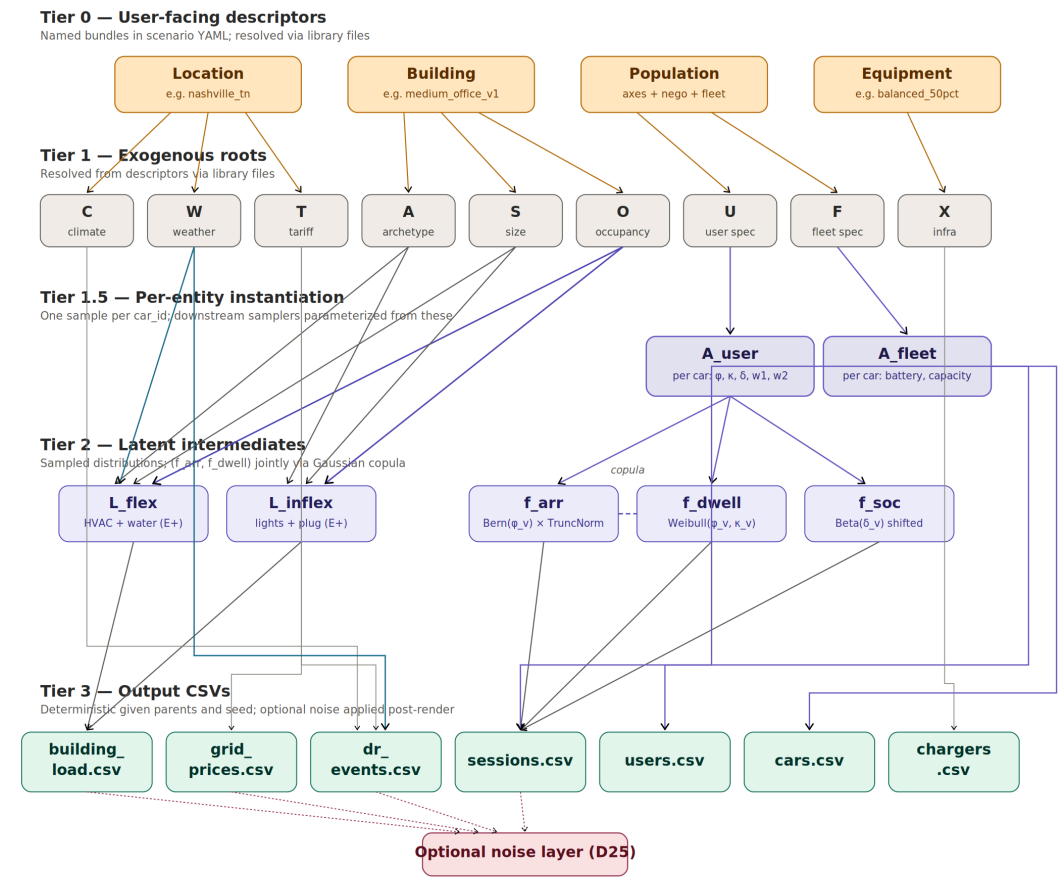
Experiment	Knobs swept	Scenarios
E1 ψ -monotonicity	population descriptors	5
E2 rate structure	tariff_type, prices, demand	3 (+S01)
E3 DR programs	dr_program, magnitude, lead	3
E4 equipment	directionality, rates	2 (+S01)
E5 CONSENT	min_depart_soc, w_mult, f_soc	4 (+S01)
E6 building	archetype, size	3 (+S01)
E8 climate $\times$ season	location $\times$ sim_window	20

4. Architecture — four tiers

v2b_syndata 4-tier architecture

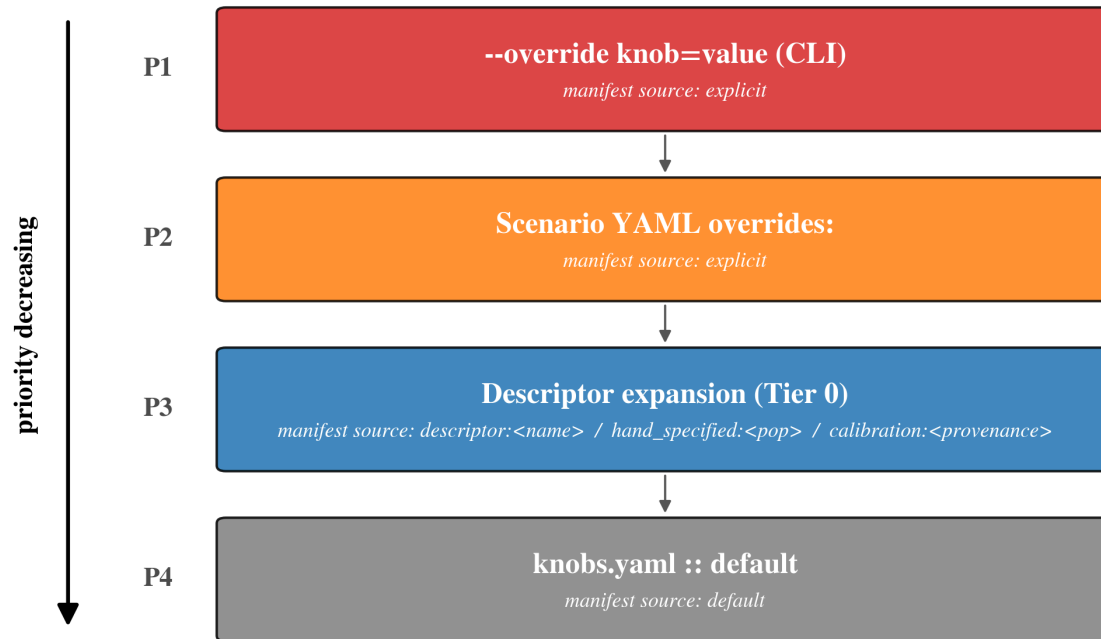


Generative DAG



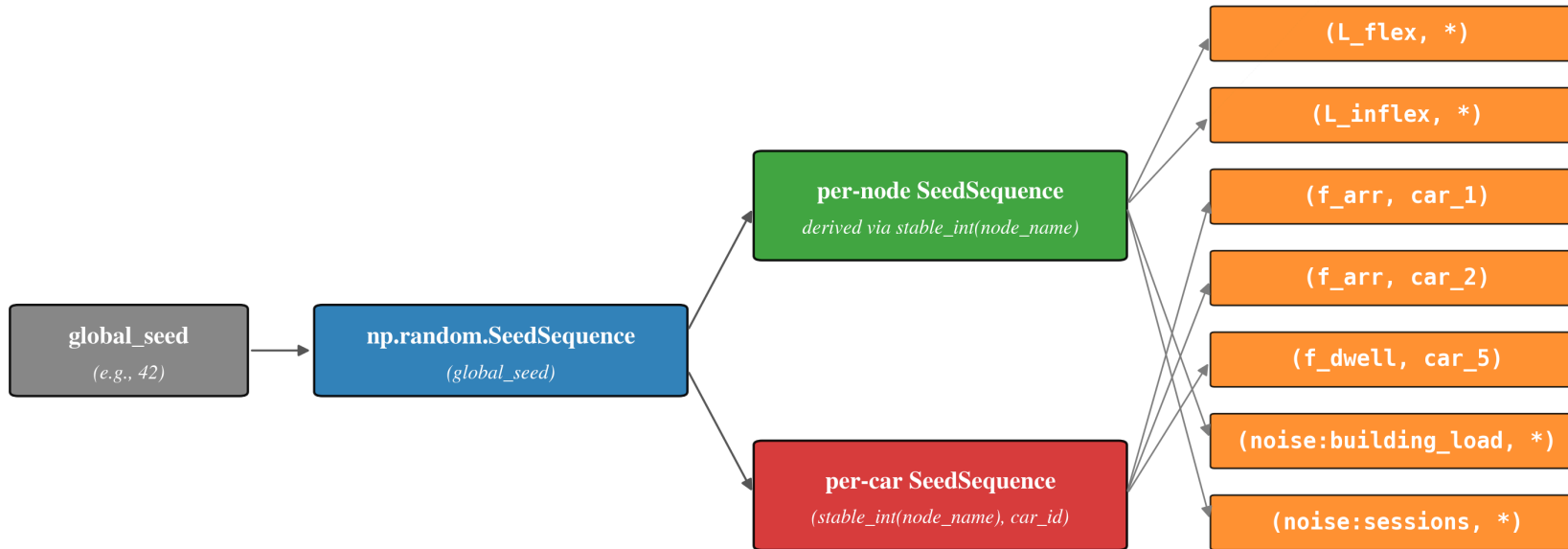
Resolution chain

Knob resolution chain (highest → lowest priority)



Seed lineage

Seeding architecture: hierarchical SeedSequence streams

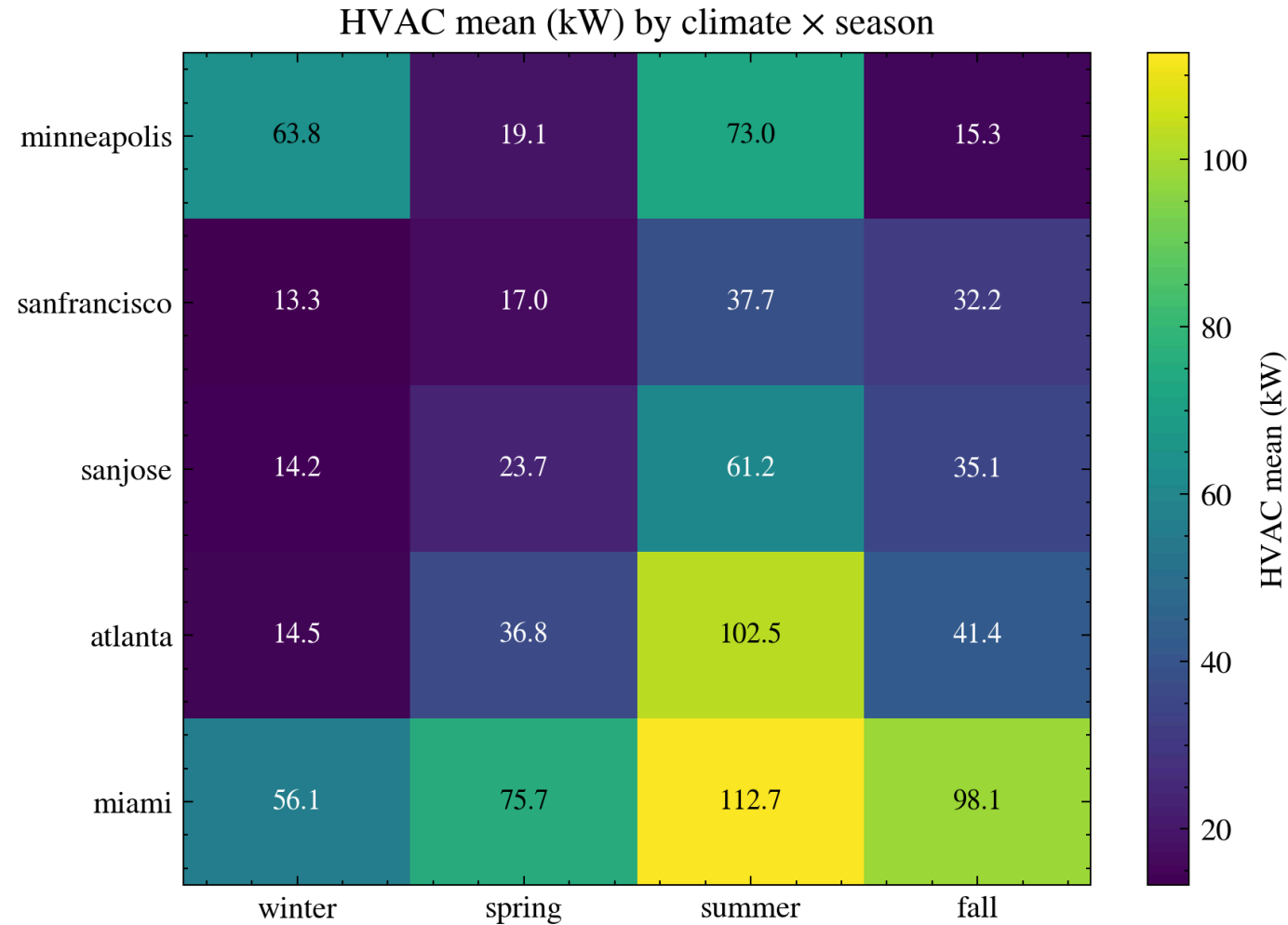


Streams are independent — adding a node doesn't shift existing streams; per-(node, car_id) derivation gives reproducible per-car randomness.

5. Data fidelity strategy

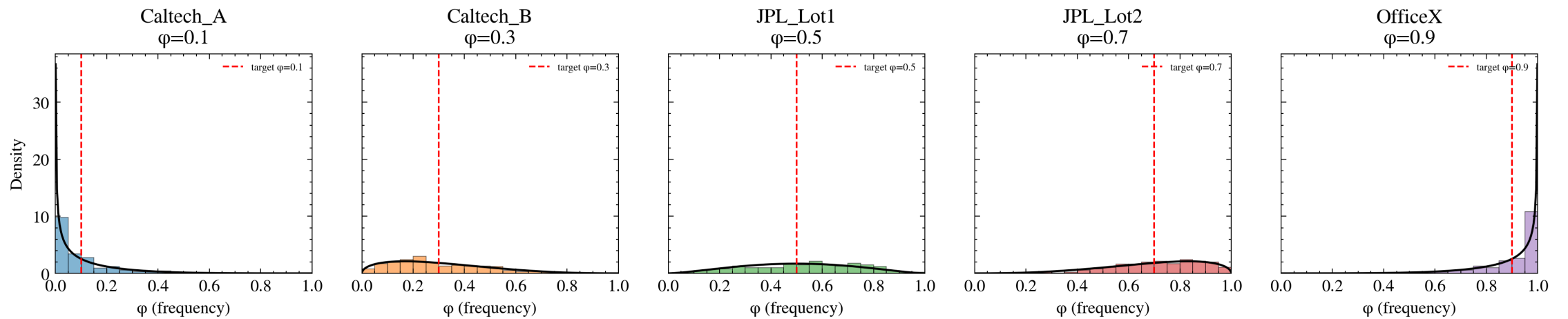
Output	Source
building_load	EnergyPlus + ASHRAE 90.1 + TMYx
sessions	Synthetic, ACN-Data calibrated
cars	Synthetic via battery_mix simplex
chargers	Synthetic
grid_prices	Rule-based from real tariffs
dr_events	Inhomogeneous Poisson (CAISO)
users	Synthetic, partially ACN-calibrated

Climate × season matrix



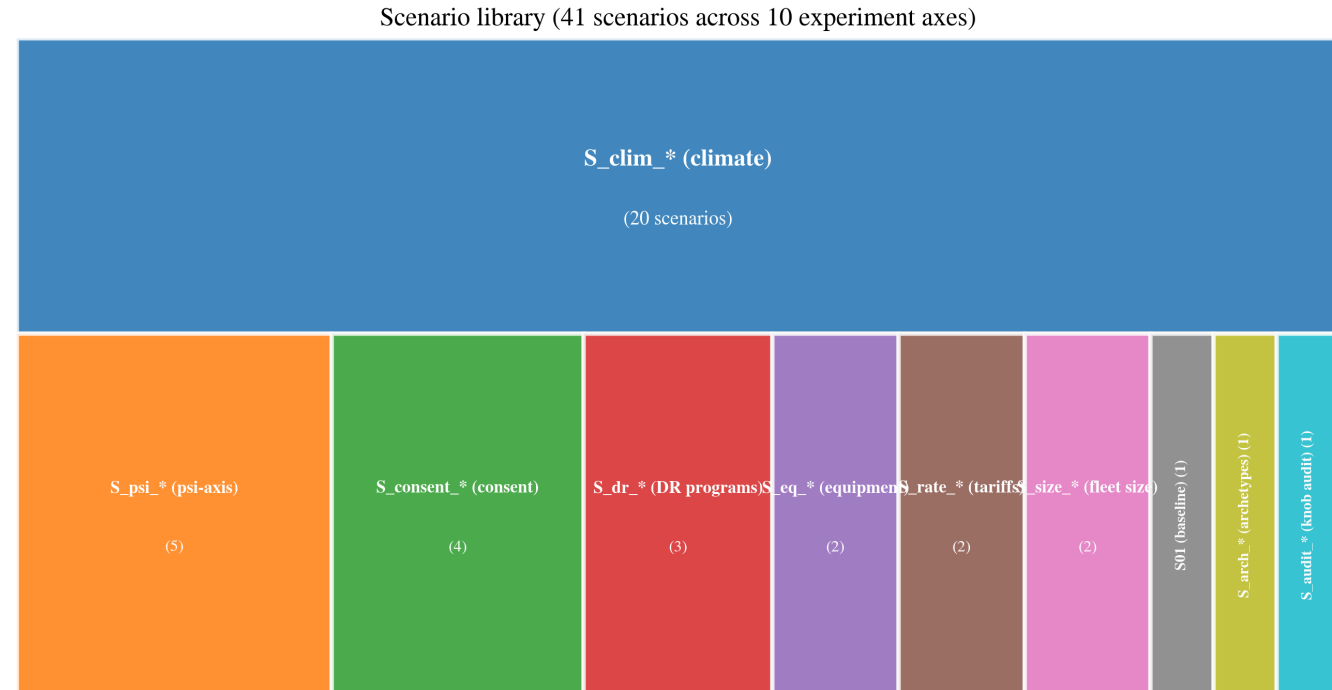
ACN-Data calibration

Synthetic illustration — ACN per-region stats not on disk: 5 ACN-anchored regions, target $\varphi = [0.1, 0.3, 0.5, 0.7, 0.9]$

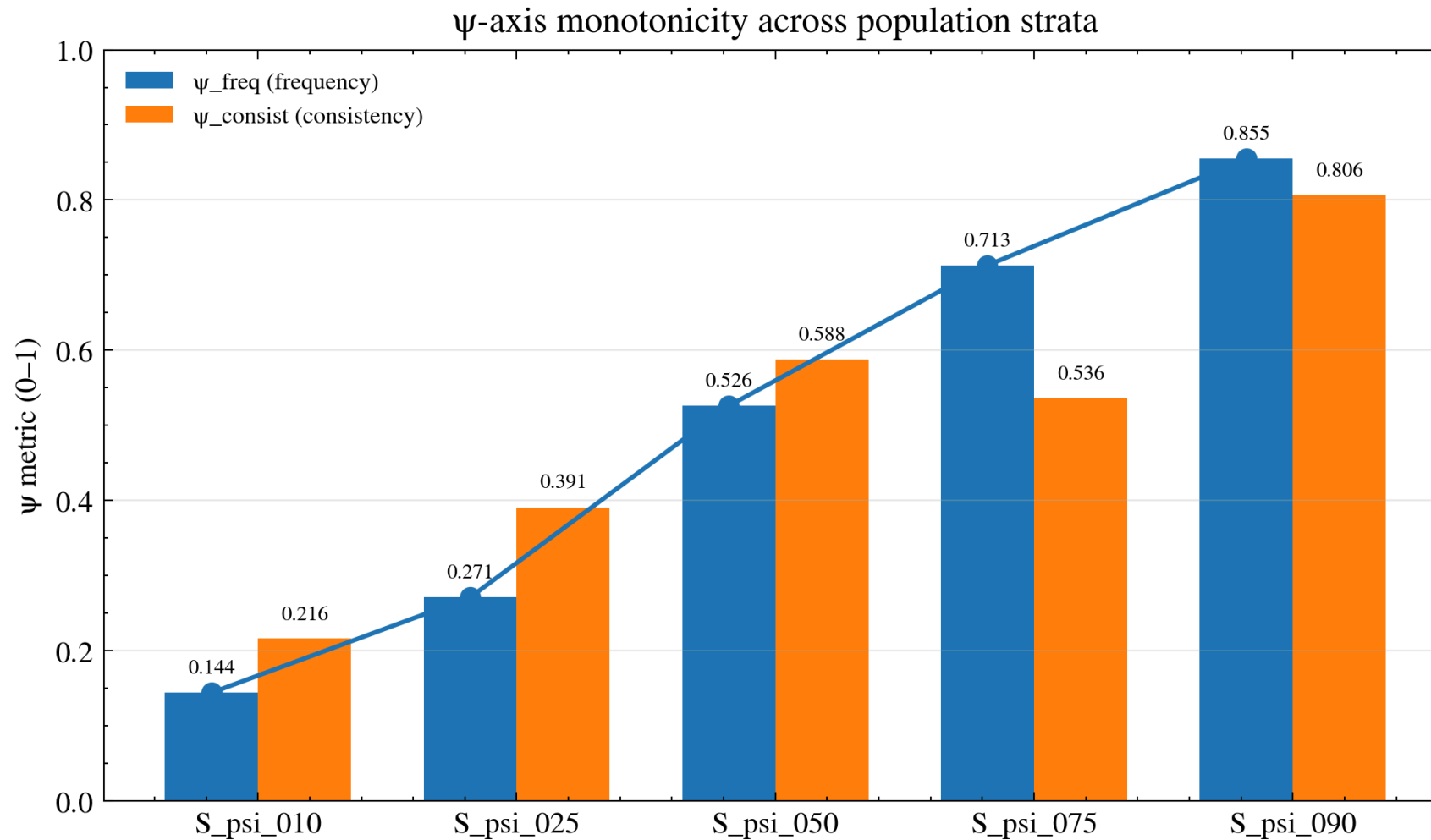


6. Scenario library

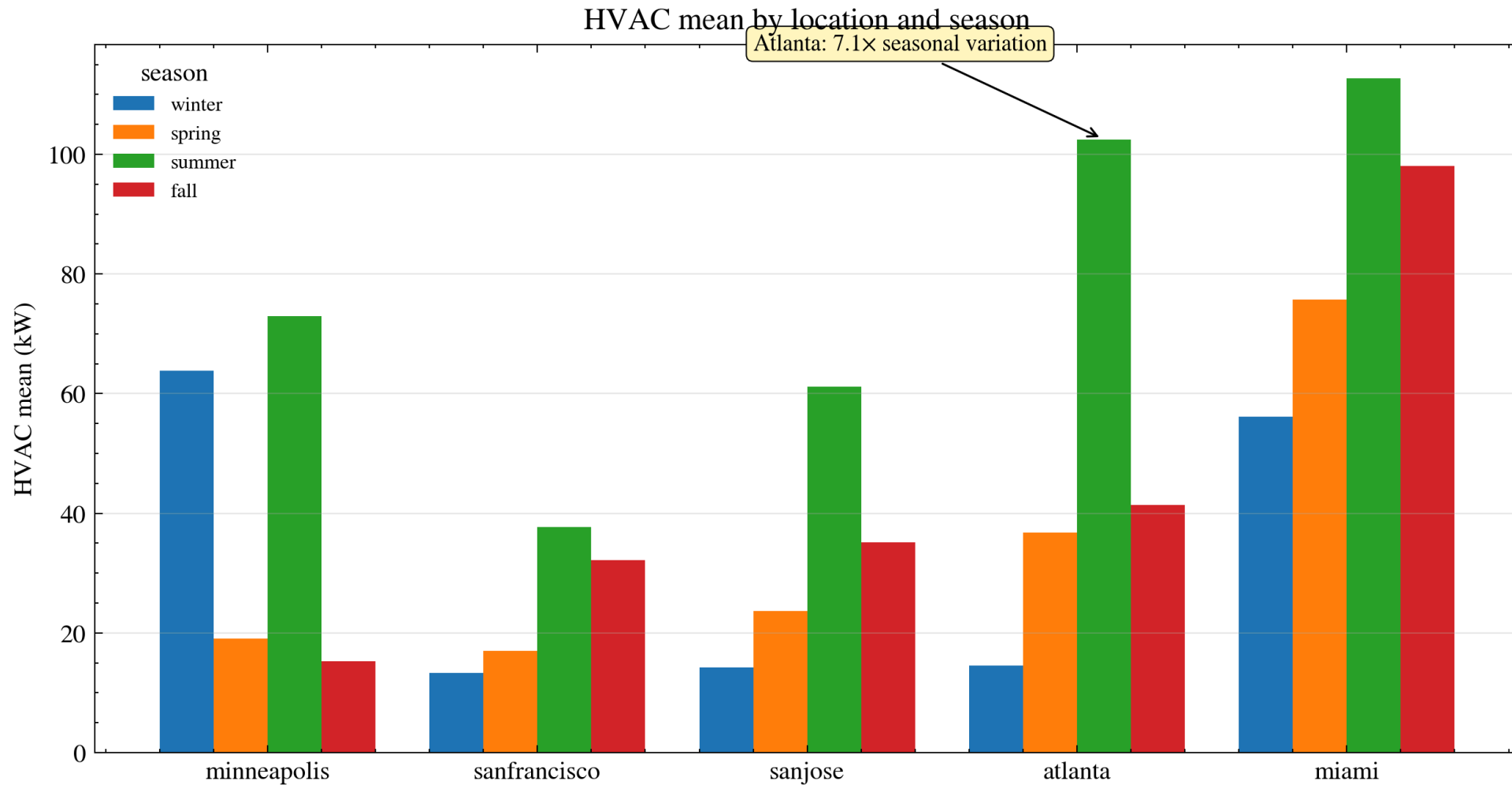
40 named scenarios cover the experiment matrix.



ψ -monotonicity



Climate divergence



7. Verification matrix

Verification coverage matrix

	Existence	Direction	Magnitude	Boundary	Pairwise	Determinism
EV Fleet	✓	✓	✓	✓	✓	✓
Charging Infra	✓	✓	✓	✓	✓	✓
User Behavior	✓	✓	✓	✓	✓	✓
Building Load	✓	✓	✓	✓	✓	✓
Utility Rate	✓	✓	✓	✓	✓	✓
Sim Window	✓	✓	✓	✓	✓	✓
Noise	✓	✓	✓	✓	✓	✓
Deep-channel	✓	✓	✓	✓	✓	✓

Verification summary

Stage	Probes	Result
Knob audit S1	98	67 HONORED
Knob audit S2	335	67/67 monotonic
V1 Coverage	pytest --cov	91% lines
V2 Boundary	48	bugs fixed
V3 Pairwise	50	0 sign-flip
V4 Determinism	11	11/11 bitwise

354 unit tests + 45 calibration tests — all passing.

Bugs caught

#	Bug	Severity
1	chargers bidir column-value mismatch	silent wrong metric
2	occupancy_jitter wrong variance	silent wrong metric
3	negotiation_mix metric used row_count	meaningless metric
4	C4 jitter: arrival > departure	invalid CSV
5	D6 jitter: arrival_soc > required	impossible state
6	missing <code>custom</code> noise profile	KeyError

- 12 declaration corrections in knobs.yaml.

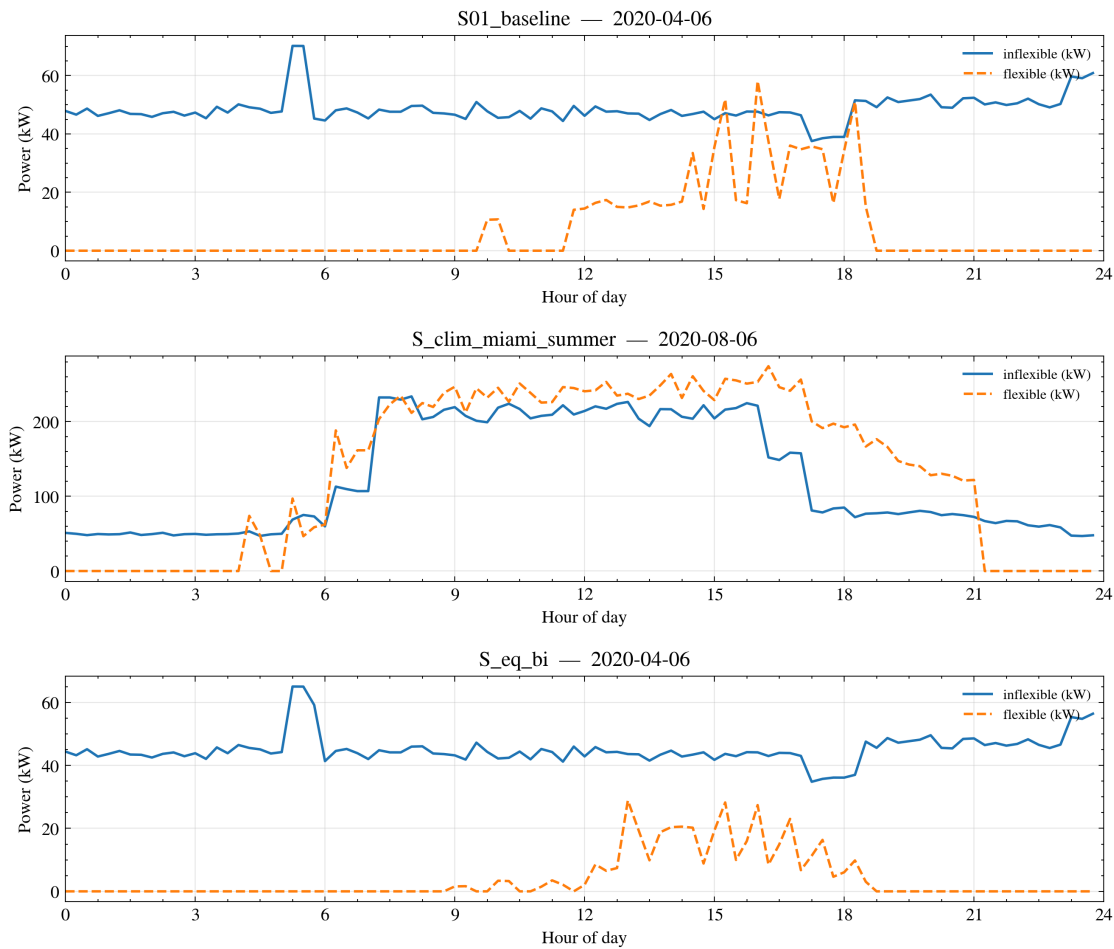
8. Working examples

Three pre-generated scenarios:

- `S01_baseline` — Nashville mid-season
- `S_clim_miami_summer` — same population, Miami / July
- `S_eq_bi` — baseline + bidirectional chargers

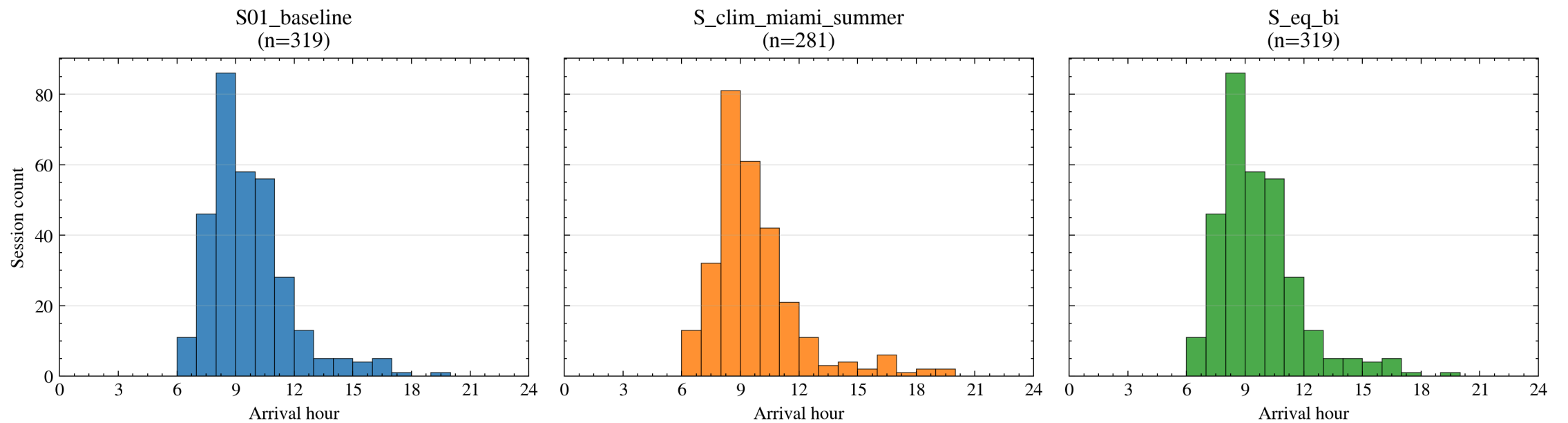
Load profiles compared

Building load profiles — single weekday across three scenarios

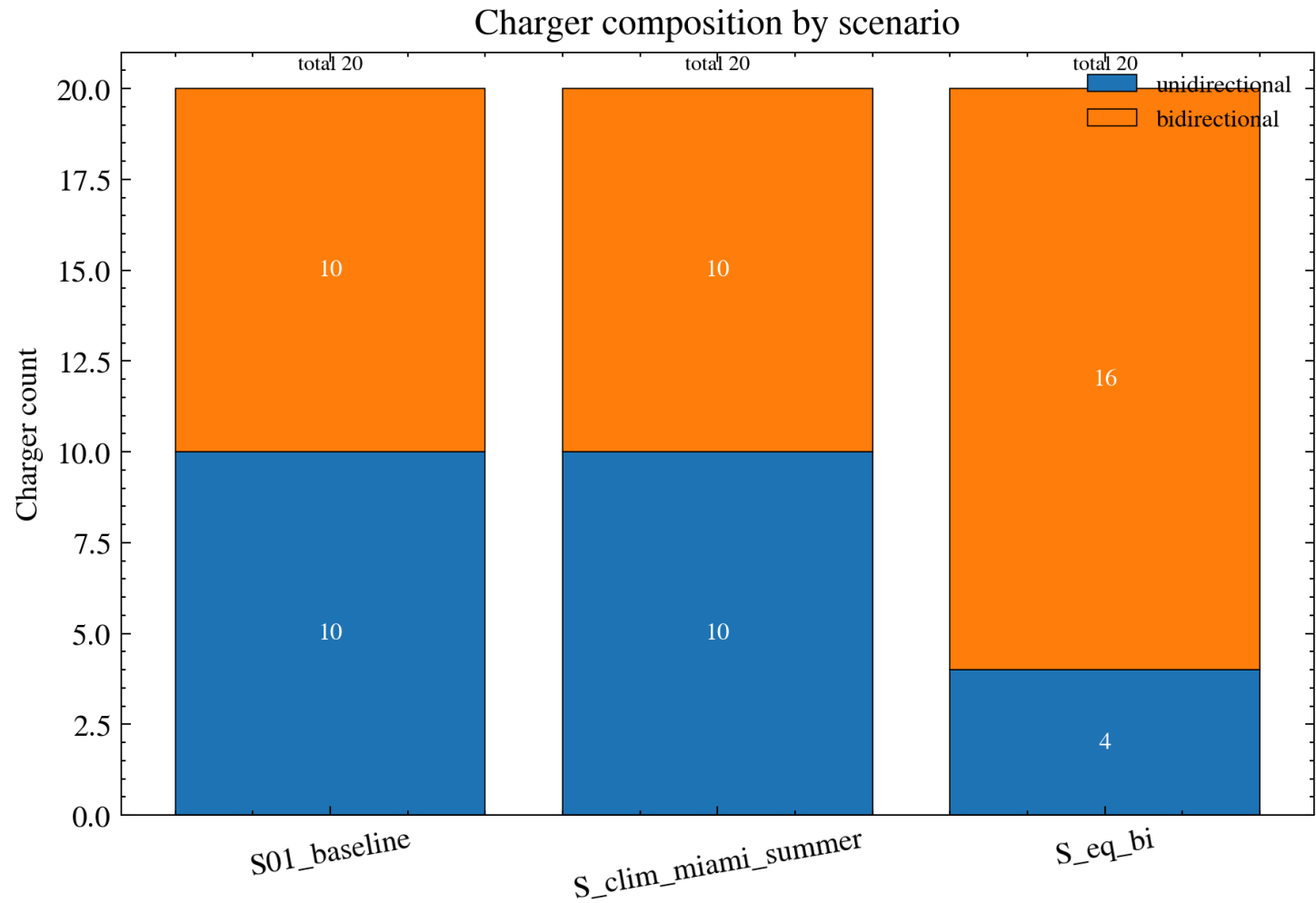


Session arrivals

Session arrival distributions (bin width = 1 hour)



Charger mixes



Knob differences

Knob resolution diff across example scenarios

Yellow cells differ from S01_baseline

Knob	S01_baseline	S_clim_miami_summer	S_eq_bi
Tariff type	TOU	DR	TOU
DR program	none	ELRP	none
TMYx station	USA_TN_Nashville.Intl.AP.7232...	USA_FL_Miami.Natl.Hurricane.C...	USA_CA_San.Jose-Mineta.Intl.A...
Sim window start	<default>	2020-08-03	<default>
Bidirectional fraction	0.50	0.50	0.80
EV fleet count	20	20	20

9. Scope and limits

Verified	Not verified
Pipeline correctness	ACN-Data generalization (research)
Reproducibility	Cross-platform binary identity
Knob effects	3+-way knob interactions
Invariant survival under noise	Real-V2B realism (research)
Determinism	Annual scale

Closing

- `OVERVIEW.md` for the narrative.
- `notebooks/exploration.ipynb` for the CSVs.
- `data/example_scenarios/` for the three working examples.
- Live frontend coming separately.